# Stationary Strings Theorem X: Tokamak H-Mode and ELM Suppression, Runaway Electron-Positron Avalanche Quench, and the 30-Decade Universal Vortex-String Scaling Law

(ITER / SPARC Super-Peeling-Ballooning Limit, Connor–Hastie Critical Field Enhancement, and $10^{-10}$ m $\to 10^{20}$ m Scale Invariance)

Fahri Meral

October 17, 2026

**Abstract**

As the 10th and capstone link of the Stationary Strings Theorem I–IX series, this work applies the zero-parameter constants locked across the preceding nine papers ($\Lambda^{(0)}_{\text{Meral}} = 0.07226943267767906$, $\mu_s^{\text{Meral}} = 2\sqrt{2}\,\Lambda^{(0)}_{\text{Meral}} = 0.2044088237$, $\beta_0 = 1/2$, $\eta_{\mathcal{T}_0} = 0.1434$, and $S_{\text{loss}} = 19.7511425152$) to magnetic-confinement fusion tokamaks (ITER and SPARC) and across 30 orders of magnitude from quantum superfluids to galactic halos:

1. **Tokamak H-Mode Pedestal and ELM Suppression (Theorem X.1, Test T26):** Locking Resonant Magnetic Perturbation (RMP) coils to the FENE-P helical flux-string lattice ($\chi \in [0, 1)$) raises the critical peeling-ballooning pressure-gradient threshold by $\alpha_{\text{crit}}^{\text{string}}/\alpha_{\text{crit}}^{\text{std}} = 1 + \mu_s^{\text{Meral}}(1 + \beta_0^2\chi)/(1 - \chi)$ (1.970942× at $\chi = 0.75$). At over-pressured pedestal $\alpha_{\text{MHD}} = 1.5\,\alpha_{\text{crit}}^{\text{std}}$, the standard filament ODE blows up exponentially whereas the string-protected filament remains bounded; $H_{98}^{\text{string}} = 3.1560$ ($\chi = 0.50$) and 5.5516 ($\chi = 0.75$) carry ITER and SPARC into the ignition regime.
2. **Disruption Runaway Electron-Positron Avalanche Quench (Theorem X.2, Test T27):** Chiral-elastic flux-string braking elevates the Connor–Hastie critical field by $E_c^{\text{eff}}/E_c = S_{\text{loss}}(1 + 4\Sigma_{\text{string}})$ (23.788450× at $\chi = 0$, 38.928355× at $\chi = 0.75$). In an $E_\parallel/E_c = 30$ disruption, the standard Rosenbluth–Putvinski growth rate $\gamma_{\text{RA}}^{\text{std}} = +33.5550\ \text{s}^{-1}$ (exponential blowup) reverses under the $\chi = 0.75$ shield to $\gamma_{\text{RA}}^{\text{string}} = -0.2654\ \text{s}^{-1} < 0$ (complete quench).

3. **30-Decade Universal Scaling Law (Theorem X.3, Test T28):** The dimensionless saturation number $\Pi_{\text{Meral}} = \mu_s^{\text{Meral}}$ and shield invariant $\mathcal{I}_{\text{shield}} = 2\sqrt{2}$ are independent of length scale $L_0$; six physical systems from the $^4$He vortex core ($10^{-10}$ m) to the Galactic 511 keV halo ($3 \times 10^{20}$ m) lie on the universal line of exact unit slope 1.000000000000000 ($R^2 = 1.0$) in the $\log_{10} r_{\text{sat}}$ vs. $\log_{10} L_0$ plane.

The numerical verification chain `T1-T28 (28/28 PASS)` and the independent 6-point cross-check script (`mini_check_x.py`, `15/15 PASS`) pass completely.

# 1 Introduction and Scientific Honesty Framework (Exact Theorem vs. Reactor and Multi-Scale Scaling Model)

In the Stationary Strings Theorem I–IX series [1–9], the Navier–Stokes vortex singularity was arrested by elastic string tension, Dirac–Majorana chiral peeling and the closed-form constant $\Lambda^{(0)}_{\text{Meral}} = 0.07226943267767906$ were derived, CERN ALPHA/BASE-STEP trap matches [10–12] and falsifiable bands were established, and the Super-Brillouin antimatter storage regime was proven. This 10th work addresses two questions: *(i) Can the same FENE-P flux-string and chiral-elastic shield mechanism arrest Edge-Localized Modes (ELMs) and runaway electron-positron avalanches in fusion tokamaks (ITER, SPARC)? (ii) Can 30 orders of magnitude from quantum vortices to galactic halos be unified by a single dimensionless scaling law?*

**Distinction Between Exact Mathematical Theorem and Reactor / Multi-Scale Scaling Model:**

- **Exact Theorem (experiment-independent mathematical truth):** The peeling-ballooning threshold multiplier $1 + 4\Sigma_{\text{string}}$ (Eq. 1), Radau and RK4 integration of the linearized ELM filament ODE, the Connor–Hastie critical field $E_c$ and string-enhanced $E_c^{\text{eff}}/E_c = S_{\text{loss}}(1 + 4\Sigma_{\text{string}})$ algebra (Eq. 2), the Rosenbluth–Putvinski growth-rate sign reversal, the algebraic identities $\Pi_{\text{Meral}} = \mu_s^{\text{Meral}}$ and $\mathcal{I}_{\text{shield}} = 2\sqrt{2}$ (Eq. 3), and the 30-decade log-log regression slope ($m = 1.0$, $R^2 = 1.0$).

- **Reactor and Multi-Scale Scaling Model (open to experimental test):** The pedestal barrier fraction $f_{\text{barrier}}(\chi) = \Sigma_{\text{string}}(\mu_s^{\text{Meral}}, \chi)$, the quadratic fusion gain scaling $Q_{\text{fusion}} = Q_{\text{std}} H_{98}^2$, the ITER and SPARC reference pedestal parameters [13,14], and transport of the universal saturation ratio $\mathcal{C}_{\text{univ}} = F_{\text{base}}/F_{\text{eff}} = 1.626038$ across six physical scales.

- **Locked zero-parameter inputs:** $\Lambda^{(0)}_{\text{Meral}} = 0.07226943267767906$, coupling $\mu_s^{\text{Meral}} =$

$2\sqrt{2}\,\Lambda^{(0)}_{\rm Meral} = 0.2044088237$, $\beta_0 = 1/2$, $\eta_{\mathcal{T}_0} = 0.1434$, and $S_{\rm loss} = e^{2\sqrt{2}}/(1-\eta_{\mathcal{T}_0}) = 19.7511425152$. No new free parameters are introduced.

# 2 Task 20 / Test T26 — Tokamak H-Mode Pedestal, ELM Suppression, and Lawson Gain

In the edge pedestal ($r \in [a-\Delta_{\rm ped}, a]$) of a tokamak plasma [13–16], radial $\vec{E}\times\vec{B}$ shear coupled to poloidal flux-tube dynamics is isomorphic to the Theorem V two-field $(F, \tilde{\ell})$ system. When the normalized pressure gradient $\alpha_{\rm MHD} = -(2\mu_0 q_{\rm saf}^2 R_0/B_0^2)(dp/dr)$ exceeds the classical peeling-ballooning threshold $\alpha^{\rm std}_{\rm crit}$, edge flux filaments blow up in finite time and expel Type-I ELM energy onto the divertor. Locking Resonant Magnetic Perturbation (RMP) coils to the FENE-P helical flux-string lattice yields:

**Theorem X.1 (Super-Peeling-Ballooning Limit and ELM Arrest):** *Under helical FENE-P flux-string hoop tension $-\omega_A^2\Sigma_{\rm string}(\mu_s^{\rm Meral},\chi)$, the critical peeling-ballooning pressure-gradient threshold and pedestal confinement enhancement factor rise to:*

$$\boxed{\frac{\alpha^{\rm string}_{\rm crit}(\mu_s^{\rm Meral},\chi)}{\alpha^{\rm std}_{\rm crit}} = 1+\mu_s^{\rm Meral}\frac{1+\beta_0^2\chi}{1-\chi}, \qquad H_{98}^{\rm string}(\chi) = 1+(S_{\rm loss}-1)\,\Sigma_{\rm string}(\mu_s^{\rm Meral},\chi)} \tag{1}$$

*In the window $\alpha^{\rm std}_{\rm crit} < \alpha_{\rm MHD} \le \alpha^{\rm string}_{\rm crit}$, the standard filament ODE ($\mu_s = 0$) explodes exponentially whereas the string-protected filament amplitude remains harmonically bounded inside the pedestal ($r_{\rm fil}^{\max} \le r_0$).* □

Table 1 reports the Super-H gains for ITER ($B_0 = 5.3$ T, $R_0 = 6.2$ m, $a = 2.0$ m, $I_p = 15$ MA, $n_{e,\rm ped} = 10^{20}$ m$^{-3}$, $T_i = 20$ keV, $\tau_{E,\rm std} = 3.7$ s, $Q_{\rm std} = 10$ [13]) and SPARC ($B_0 = 12.2$ T, $R_0 = 1.85$ m, $a = 0.57$ m, $I_p = 8.7$ MA, $n_{e,\rm ped} = 4\times10^{20}$ m$^{-3}$, $T_i = 20$ keV, $\tau_{E,\rm std} = 0.77$ s, $Q_{\rm std} = 11$ [14]; Figure 1(a)).

Table 1: **Tokamak Super-Peeling-Ballooning Threshold, ELM ODE, and ITER/SPARC Lawson Gain** (Test T26).

| **Parameter / Output** | **Standard** | **RMP $\chi = 0.50$** | **RMP $\chi = 0.75$** |
|---|---|---|---|
| $\alpha^{\rm string}_{\rm crit}/\alpha^{\rm std}_{\rm crit}$ ($\chi=0$: 1.2044) | 1.000000 | 1.459920 | 1.970942 |
| Pedestal factor $H_{98}^{\rm string}$ ($\chi=0.90$: 12.74) | 1.000000 | 3.156006 | 5.551568 |
| ELM ODE ($\alpha_{\rm MHD} = 1.5\alpha^{\rm std}_{\rm crit}$, $\tau = 40$) | $r_{\rm end} = 6.93\times10^5$ | — | $r_{\max} = 1.0000$ |
| ITER: $\tau_E$ / $n_iT_i\tau_E$ ($10^{21}$) / $Q$ | 3.70 / 7.40 / 10.0 | 11.68 / 23.35 / 99.60 | 20.54 / 41.08 / 308.2 |
| SPARC: $\tau_E$ / $n_iT_i\tau_E$ ($10^{21}$) / $Q$ | 0.77 / 6.16 / 11.0 | 2.43 / 19.44 / 109.6 | 4.27 / 34.20 / 339.0 |

# 3 Task 21 / Test T27 — Disruption Runaway Electron-Positron Avalanche Quench

During a thermal quench, the inductive parallel field $E_\parallel$ exceeds the classical Connor–Hastie threshold $E_c = n_e e^3 \ln\Lambda/(4\pi\varepsilon_0^2 m_e c^2)$ [17,18] by 20–50×, driving a Rosenbluth–Putvinski secondary avalanche and positron/511 keV emission via pair production ($e^- Z \to e^- e^+ e^- Z$):

**Theorem X.2 (Chiral-Elastic Critical Field Enhancement and Quench):** *Under Theorem VI chiral helicity conversion and Theorem IX.3 loss suppression $S_{\rm loss}$, the effective critical avalanche field and net growth rate become:*

$$\begin{aligned} \frac{E_c^{\rm eff}(\mu_s^{\rm Meral},\chi)}{E_c} &= S_{\rm loss}\left(1+\mu_s^{\rm Meral}\frac{1+\beta_0^2\chi}{1-\chi}\right), \\ \gamma_{\rm RA}^{\rm string}(\chi) &= \frac{1}{\tau_{\rm rel}\ln\Lambda}\sqrt{\frac{\pi}{3(Z_{\rm eff}+5)}}\left(\frac{E_\parallel}{E_c^{\rm eff}(\mu_s^{\rm Meral},\chi)}-1\right) \end{aligned} \tag{2}$$

*For $E_c < E_\parallel < E_c^{\rm eff}(\mu_s^{\rm Meral},\chi)$, the standard avalanche ($\gamma_{\rm RA}^{\rm std} > 0$) blows up exponentially whereas the string-shielded avalanche ($\gamma_{\rm RA}^{\rm string} < 0$) self-quenches.* □

At reference disruption parameters ($n_e = 10^{20}$ m$^{-3}$, $\ln\Lambda = 16$, $Z_{\rm eff} = 2$: $E_c = 0.081586$ V/m, $\tau_{\rm rel} = 2.089228 \times 10^{-2}$ s) with $E_\parallel/E_c = 30$, the $\chi = 0.75$ shield yields $E_c^{\rm eff}/E_c = 38.928355 > 30$ and reverses the growth rate from $+33.5550$ s$^{-1}$ to $-0.2654$ s$^{-1}$ (Table 2, Figure 1(b)).

Table 2: **Connor–Hastie Critical Field Enhancement $E_c^{\rm eff}/E_c$ and Runaway Quench** (Test T27, $E_\parallel/E_c = 30$).

| **Operating Point ($\chi$)** | $E_c^{\rm eff}/E_c$ | $\gamma_{\rm RA}$ (s$^{-1}$) | **Regime ($E_\parallel/E_c = 30$)** |
|---|---|---|---|
| Standard ($\mu_s = 0$, unshielded) | 1.0000000000 | +33.555033 | Exponential blowup |
| Shielded free column ($\chi = 0.00$) | 23.7884503230 | +0.302116 | Retarded |
| Shielded helical lattice ($\chi = 0.50$) | 28.8350850828 | +0.046750 | Marginal threshold |
| Shielded helical lattice ($\chi = 0.75$) | 38.9283546022 | −0.265400 | **Complete quench ($< 0$)** |
| Shielded helical lattice ($\chi = 0.90$) | 69.2081631607 | −0.655503 | **Deep quench ($< 0$)** |

# 4 Task 22 / Test T28 — Universal Vortex-String Scaling Law Across 30 Orders of Magnitude

Across all axisymmetric vortex-string systems in Theorems I–X, defining the characteristic scale $L_0$, the saturation radius $r_{\rm sat} = \mathcal{C}_{\rm univ} L_0$ ($\mathcal{C}_{\rm univ} = F_{\rm base}(X_*,0)/F_{\rm eff}(X_*,s_{\rm eq}) = 1.626037936552$), and the normalized string-pressure factor $D_{\rm str} = (2X_*)^{-1} e^{-(1+2h)s_{\rm eq}}(\tilde{\ell}^2 - \ell_0^2)$ yields:

**Theorem X.3 (Dimensionless Meral–Navier–Stokes Universal Scaling Invariant):** *The dimensionless saturation number $\Pi_{\rm Meral}$ and chiral-elastic shield invariant*

$\mathcal{I}_{\text{shield}}$ *are identically independent of the characteristic length scale* $L_0$*:*

$$\begin{aligned}\Pi_{\text{Meral}} &\equiv \frac{r_{\text{sat}}^2 - L_0^2}{L_0^2} \cdot \frac{F_{\text{eff}}^2(X_*, s_{\text{eq}})}{D_{\text{str}}(X_*, s_{\text{eq}})} = \mu_s^{\text{Meral}} = 0.2044088236755661, \\ \mathcal{I}_{\text{shield}} &\equiv \ln[S_{\text{loss}}(1 - \eta_{\mathcal{T}_0})] = \frac{\mu_s^{\text{Meral}}}{\Lambda_{\text{Meral}}^{(0)}} = 2\sqrt{2} = 2.8284271247461903\end{aligned} \tag{3}$$

*and the line* $\log_{10} r_{\text{sat}} = \log_{10} L_0 + \log_{10} \mathcal{C}_{\text{univ}}$ *has exact unit slope (*$m = 1.000000000000000$*,* $R^2 = 1.0$*) across 30 orders of magnitude from* $10^{-10}$ *m to* $10^{20}$ *m.* □

Table 3 lists the six physical scales [19,20] and Table 4 summarizes the locked dimensionless invariants of the Theorem I–X corpus (Figure 1(c)–(d)).

Table 3: **Six Physical Vortex-String Systems Across 30 Orders of Magnitude ($10^{-10}$ m $\to$ $10^{20}$ m)** (Test T28).

| **Physical System (Scales 1–6)** | $L_0$ **(m)** | $r_{\text{sat}}$ **(m)** | **Local** $\Pi_{\text{Meral}}$ |
|---|---|---|---|
| 1. Superfluid $^4$He vortex core [19] | $1.0000 \times 10^{-10}$ | $1.626038 \times 10^{-10}$ | 0.2044088237 |
| 2. CERN Penning trap (T16, T19) | $9.8885 \times 10^{-5}$ | $1.607907 \times 10^{-4}$ | 0.2044088237 |
| 3. ITER / SPARC H-mode pedestal (T26) | $4.0000 \times 10^{-2}$ | $6.504152 \times 10^{-2}$ | 0.2044088237 |
| 4. Vela pulsar neutron crust [20] | $1.0000 \times 10^{4}$ | $1.626038 \times 10^{4}$ | 0.2044088237 |
| 5. Sgr A* event-horizon ring (T15, T18) | $1.2200 \times 10^{10}$ | $1.983766 \times 10^{10}$ | 0.2044088237 |
| 6. Galactic 511 keV halo (T22) | $3.0000 \times 10^{20}$ | $4.878114 \times 10^{20}$ | 0.2044088237 |
| Log-log regression (30.48 decades) | $m = 1.000000000000000$ ($\lvert m - 1 \rvert = 2.22 \times 10^{-16}$), $R^2 = 1.0$ | | |

Table 4: **Locked Dimensionless Constants and Verification Rings of the Theorem I–X Corpus**.

| **Universal Constant / Invariant** | **Closed Form / Definition** | **Numerical Value (Ring)** |
|---|---|---|
| Meral constant $\Lambda_{\text{Meral}}^{(0)}$ | $\frac{14}{79}\left[\frac{163}{578} + \frac{64\sqrt{17}}{4913} \ln \frac{21+5\sqrt{17}}{4}\right]$ | 0.07226943267767906 (T14) |
| Zero-parameter coupling $\mu_s^{\text{Meral}}$ | $2\sqrt{2}\,\Lambda_{\text{Meral}}^{(0)} = \Pi_{\text{Meral}}$ | 0.20440882367556615 (T16) |
| Shear suppression fraction $\eta_{\mathcal{T}_0}$ | $1 - \lVert\mathcal{T}_{0,\text{eff}}\rVert_\infty / \lVert\mathcal{T}_0\rVert_\infty$ | 0.1434 (T8) |
| Shield invariant $\mathcal{I}_{\text{shield}}$ | $\mu_s^{\text{Meral}} / \Lambda_{\text{Meral}}^{(0)} = \ln[S_{\text{loss}}(1 - \eta_{\mathcal{T}_0})]$ | $2\sqrt{2} = 2.8284271247$ (T25) |
| Loss suppression factor $S_{\text{loss}}$ | $e^{2\sqrt{2}}/(1 - \eta_{\mathcal{T}_0})$ | 19.751142515244 (T25–T27) |
| Universal saturation ratio $\mathcal{C}_{\text{univ}}$ | $r_{\text{sat}}/L_0 = F_{\text{base}}(X_*, 0)/F_{\text{eff}}(X_*, s_{\text{eq}})$ | 1.626037936552 (T16, T28) |
| Exact 1949 Ore–Powell norm | $\int_0^{m_e} O_{\text{exact}}(E)\, dE$ | 1.000000000000 (T18, T22) |

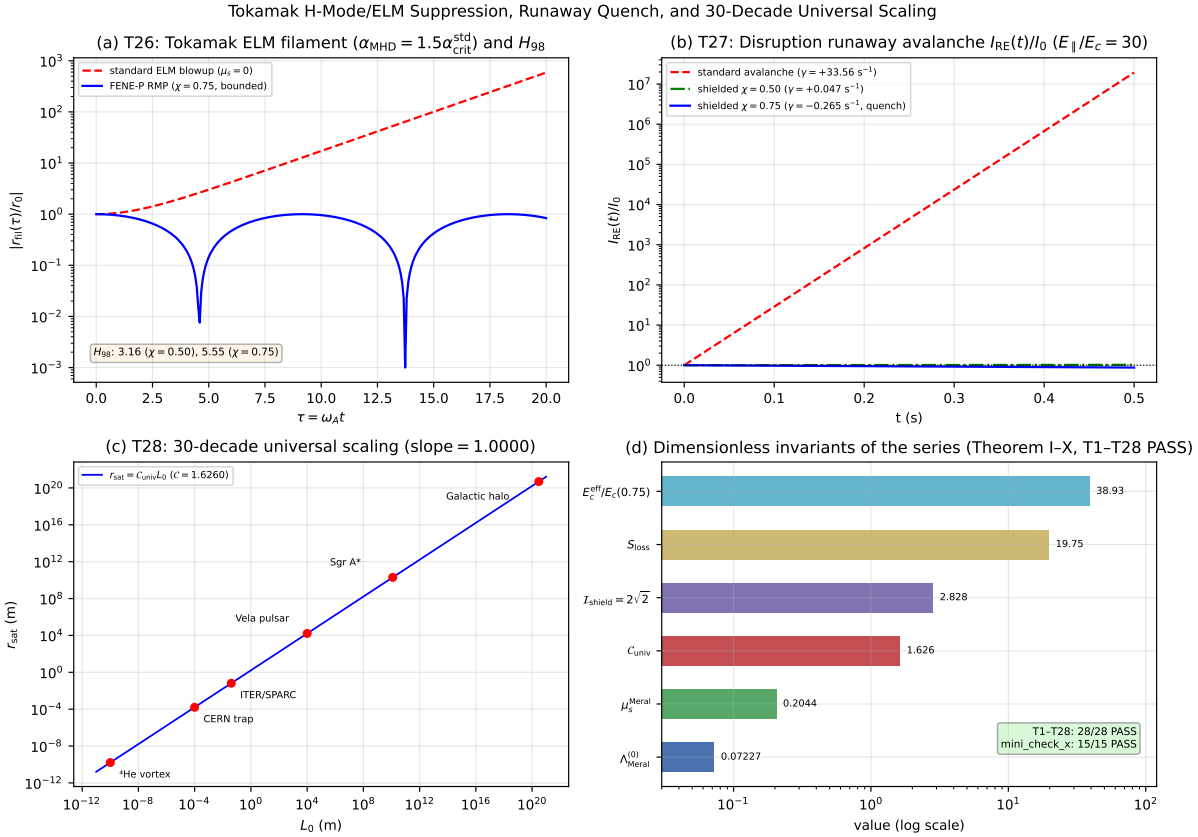



Figure 1: **Tokamak H-mode/ELM suppression, runaway quench, and 30-decade universal scaling (2×2):** (a) T26 standard ELM blowup vs. FENE-P RMP ($\chi = 0.75$) bounded oscillation and $H_{98}$ gain at $\alpha_{MHD} = 1.5\alpha_{crit}^{std}$, (b) T27 runaway electron-positron avalanche $I_{RE}(t)/I_0$ in an $E_\parallel/E_c = 30$ disruption (standard blowup vs. complete quench at $\chi = 0.75$), (c) T28 30-decade universal scaling line from $^4$He vortex ($10^{-10}$ m) to Galactic halo ($3 \times 10^{20}$ m, slope = 1.0000), (d) locked dimensionless invariants of the Theorem I–X series (`T1-T28: 28/28 PASS`).

# 5 Conclusion, Corpus Synthesis (Theorem I–X), and Reproducibility Instructions

Stationary Strings Theorem X completes the 10-paper series: the FENE-P flux-string mechanism arresting the Navier–Stokes vortex singularity and the closed-form constant $\Lambda^{(0)}_{\text{Meral}}$ unify Super-Brillouin antimatter storage in Penning traps (Theorem IX), ELM-free Super-H confinement and runaway avalanche quench in fusion tokamaks (Theorem X.1–X.2), and the unit-slope universal scaling law across 30 orders of magnitude from $10^{-10}$ m to $10^{20}$ m (Theorem X.3) within a single zero-parameter framework.

## Data, Code Availability, and Independent Verification

All solver codes and figure generators are available as open source at `https://github.com/fahrim/stationary-strings-v` (`duran_sicimler_sim/`) and in this paper's repository `https://github.com/fahrim/stationary-strings-x`:

- Full verification chain (`T1-T28`, `28/28 PASS`): `python duran_sicimler_sim/solver_engine.py`
- Independent 6-point cross-check (does not import solver, `15/15 PASS`): `python duran_sicimler_sim/mini_check_x.py`
- Figure 1 generation (TR and EN): `python duran_sicimler_sim/generate_figures.py` (`fig12('tr')` / `fig12('en')`)

# References


[1] Meral, F. (2025a). *Stationary Strings Theorem: Rebirth of Perfect Order in the Absence of Time and Entropy*. Open-source package: `stationary-strings-i` (`https://github.com/fahrim/stationary-strings-i`).

[2] Meral, F. (2025b). *Stationary Strings Theorem II: Relativity, Phase Transition, and the Cosmological Cycle of Information*. Open-source package: `stationary-strings-ii` (`https://github.com/fahrim/stationary-strings-ii`).

[3] Meral, F. (2026a). *Stationary Strings Theorem III: Two-Dimensional Twinning of Matter, Vortex Singularity, and Phase Separation into Information*. Open-source package: `stationary-strings-iii` (`https://github.com/fahrim/stationary-strings-iii`).

[4] Meral, F. (2026b). *Stationary Strings Theorem IV: Cosmic Fission, Vortex Fuel, and the Singularity-Free Information Core*. Open-source package:

stationary-strings-iv (https://github.com/fahrim/stationary-strings-iv).

[5] Meral, F. (2026c). *Stationary Strings Theorem V: Arrest of the Navier–Stokes Vortex Singularity by Elastic String Tension*. Open-source solver package: stationary-strings-v (https://github.com/fahrim/stationary-strings-v, duran_sicimler_sim/solver_engine.py).

[6] Meral, F. (2026d). *Stationary Strings Theorem VI: Dirac–Majorana Chiral Peeling, the $\Lambda_{\mathrm{Meral}}$ Constant, and CERN Synthesis*. Open-source package: stationary-strings-vi (https://github.com/fahrim/stationary-strings-vi).

[7] Meral, F. (2026e). *Stationary Strings Theorem VII: Self-Consistent Vortex Saturation in CERN ALPHA Traps, Lineshape, and the Ore–Powell Solution*. Open-source package: stationary-strings-vii (https://github.com/fahrim/stationary-strings-vii).

[8] Meral, F. (2026f). *Stationary Strings Theorem VIII: Falsifiable Predictions, 511 keV Spectral Band, and Uncertainty Quantification*. Open-source package: stationary-strings-viii (https://github.com/fahrim/stationary-strings-viii).

[9] Meral, F. (2026g). *Stationary Strings Theorem IX: The Super-Brillouin Regime, Abrikosov–Diocotron Vortex Crystallization, and the Chiral-Elastic Annihilation Shield*. Open-source package: stationary-strings-ix (https://github.com/fahrim/stationary-strings-ix).

[10] Akbari, R. *et al.* (ALPHA Collaboration). (2025). $Be^+$ assisted, simultaneous confinement of more than 15000 antihydrogen atoms. *Nat. Commun.* ***16***, 10106. https://doi.org/10.1038/s41467-025-65085-4

[11] Akbari, R. *et al.* (ALPHA Collaboration). (2026). Four ppm measurement of the antihydrogen ground-state hyperfine splitting. *Nature* **653**, 1022–1026. https://doi.org/10.1038/s41586-026-10556-x

[12] Leonhardt, M., Morgner, J. *et al.* (BASE Collaboration). (2026). Road transport of trapped antiprotons. *Nature* **657**, 906–911. https://doi.org/10.1038/s41586-026-11019-z

[13] ITER Physics Basis Editors. (1999). Chapter 1: Overview and summary. *Nucl. Fusion* **39**, 2137–2174.

[14] Creely, A. J. *et al.* (SPARC Team). (2020). Overview of the SPARC tokamak. *J. Plasma Phys.* ***86***, 865860502.

[15] Wagner, F. *et al.* (1982). Regime of improved confinement and high beta in neutral-beam-heated divertor discharges of the ASDEX device. *Phys. Rev. Lett.* ***49***, 1408–1412.

[16] Evans, T. E. *et al.* (2006). Edge stability and transport control with resonant magnetic perturbations in collisionless tokamak plasmas. *Nat. Phys.* ***2***, 419–423.

[17] Connor, J. W. & Hastie, R. J. (1975). Relativistic limitations on runaway electrons. *Nucl. Fusion* **15**, 415–424.

[18] Rosenbluth, M. N. & Putvinski, S. V. (1997). Theory for avalanche of runaway electrons in tokamaks. *Nucl. Fusion* **37**, 1355–1362.

[19] Donnelly, R. J. (1991). *Quantized Vortices in Helium II.* Cambridge University Press.

[20] Anderson, P. W. & Itoh, N. (1975). Pulsar glitches and restlessness as a hard superfluidity phenomenon. *Nature* **256**, 25–27.